

# Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

**Benchmark:** MMLU | **Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

MMLU applies a uniform four-class MCQ schema (A/B/C/D) across all 57 tasks [Q23, Q35], with no mechanism to encode legitimate regional divergence in correct answers. The deployment explicitly anticipates such divergence — particularly for Bangladesh–India water-sharing questions [elicitation A3] where active 2025 geopolitical context (Ganges Treaty renewal, unresolved Teesta dispute, Kushiya agreement) [WEB-1, WEB-2, WEB-4, WEB-21] would yield nationally divergent correct answers. The moral\_scenarios config (500 items) explicitly anchors correctness to 'ordinary moral standards in the US as of 2020' as a constitutive framing across every item [DATASET-D1, D12], structurally precluding pluralism. The four-option forced-choice schema also cannot capture practice differences between delta flood-cycle planning and Deccan dry-land cropping. Models perform especially poorly on socially-important subjects (morality, law) [Q6, Q86] and on procedural/calculation tasks [Q14, Q72] — precisely the categories most relevant to agricultural decision-making. OO is HIGH-priority for this deployment, and the schema rigidity is a structural validity violation.

| ID     | Evidence                                                                                                                                                                                                                                                                              |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q6     | "Worse, they still have near-random accuracy on some socially important subjects such as morality and law." (p.1)                                                                                                                                                                     |
| Q14    | "The tasks with near-random accuracy include calculation-heavy subjects such as physics and mathematics and subjects related to human values such as law and morality." (p.2)                                                                                                         |
| Q23    | "we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)                                                                                                                                                               |
| Q35    | "To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)                                                                                                                                                           |
| Q72    | "Models also have difficulty performing calculations, so much so that they exhibit poor performance on Elementary Mathematics and many other STEM subjects with "plug and chug" problems." (p.8)                                                                                      |
| Q86    | "Worryingly, models also perform especially poorly on socially relevant subjects including morality and law." (p.8)                                                                                                                                                                   |
| D1     | For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, **according to ordinary moral standards in the US as of 2020**? — Question explicitly frames moral standards as US-specific, 2020 vintage                             |
| D12    | "For which of these two scenarios does the main character ... do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?" (repeated framing across all 5 examples) — Every moral_scenarios question frames correctness relative to US 2020 norms |
| WEB-1  | Active Bangladesh–India treaty renewal context — divergent correct answers likely (https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty)                                                                         |
| WEB-2  | Teesta dispute unresolved — Bangladesh and India would adjudicate water-allocation questions differently (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)                                                                                                                         |
| WEB-4  | Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ (https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696)                                                                             |
| WEB-21 | Kushiya River Agreement — recent bilateral agreement not reflected in 2021 benchmark labels (https://www.waterdiplomat.org/story/2025/08/teesta-river-politics-and-benefit-sharing-getting-yes-without-grand-bargain)                                                                 |

## Strengths

- Four-option MCQ schema is simple and reproducible for cross-model accuracy comparison [Q23, Q35]

- Difficulty levels (Elementary/High School/College/Professional) provide some stratification useful for capacity profiling [Q20]

| ID  | Evidence                                                                                                                                                       |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q20 | "Some tasks cover a subject, like psychology, but at a specific level of difficulty, such as "Elementary," "High School," "College," or "Professional."" (p.3) |
| Q23 | "we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)                                        |
| Q35 | "To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)                                    |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                          |
|------|--------------------------------------------------------------|
| OO-1 | Review output label categories for regional relevance        |
| OO-2 | Identify missing regionally specific categories              |
| OO-3 | Flag categories encoding non-regional values                 |
| OO-4 | Consider stakeholder-driven taxonomy redesign                |
| OO-5 | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** The MCQ label space (A/B/C/D) [Q23] is generic and content-agnostic; it is not regionally relevant or irrelevant per se, but it cannot represent the categorical decision boundaries needed for regional agricultural reasoning (e.g., 'optimal under delta flood regime' vs. 'optimal under Deccan dry-land regime' as parallel correct answers).

**OO-2:** Missing categories include: regional-pluralism markers (which national policy regime applies), agro-ecological zone qualifiers, and any output that would let a model express 'this depends on whether the question is from a Bangladeshi or Indian regulatory perspective' [WEB-1, WEB-2 — active treaty divergence].

**OO-3:** The moral\_scenarios subtask explicitly encodes US 2020 norms as the constitutive correctness criterion [DATASET-D1, D12], embedding non-regional values directly into the output ontology. Human\_sexuality and human\_aging similarly embed US institutional facts as universally correct outputs [DATASET-D9, D10, D11].

**OO-4:** Significant misalignment exists; stakeholder-driven taxonomy redesign would require either (a) adding parallel-answer schema for regionally-divergent items or (b) excluding the affected subtasks. The paper itself notes models 'are notably poor at modeling human (dis)approval' [Q71], compounding OO concerns on socially-situated items.

**OO-5:** The combination of forced single-correct-answer MCQ + US-anchored normative subtasks + procedural-calculation weakness [Q72] across calculation-heavy tasks (Elementary Mathematics RMS calibration error 19.4% [Q62]) creates structural validity issues for an agricultural decision-support evaluation.

| ID  | Evidence                                                                                                                                                                                                                                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q6  | "Worse, they still have near-random accuracy on some socially important subjects such as morality and law." (p.1)                                                                                                                                                   |
| Q14 | "The tasks with near-random accuracy include calculation-heavy subjects such as physics and mathematics and subjects related to human values such as law and morality." (p.2)                                                                                       |
| Q23 | "we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)                                                                                                                                             |
| Q62 | "Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%." (p.7) |
| Q71 | "They are notably poor at modeling human (dis)approval, as evident by the low performance on the Professional Law and Moral Scenarios tasks." (p.8)                                                                                                                 |
| Q72 | "Models also have difficulty performing calculations, so much so that they exhibit poor performance on Elementary Mathematics and many other STEM subjects with "plug and chug" problems." (p.8)                                                                    |
| Q86 | "Worryingly, models also perform especially poorly on socially relevant subjects including morality and law." (p.8)                                                                                                                                                 |

|       |                                                                                                                                                                                                                                                                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1    | For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, **according to ordinary moral standards in the US as of 2020**? — Question explicitly frames moral standards as US-specific, 2020 vintage                                                                                               |
| D9    | The age at which one can receive full benefits from social security" with answer "Has been increasing — US Social Security system — entirely US institutional context                                                                                                                                                                                   |
| D10   | Which of the following is the most common method of birth control for married couples in the **U.S.** — Explicitly US demographic data                                                                                                                                                                                                                  |
| D11   | From 1988 to 1990 among heterosexuals in the **US**, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US-specific behavioral survey data                                                                                                                                         |
| D12   | "For which of these two scenarios does the main character ... do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?" (repeated framing across all 5 examples) — Every moral_scenarios question frames correctness relative to US 2020 norms                                                                   |
| WEB-1 | Active Bangladesh–India treaty renewal context — divergent correct answers likely ( <a href="https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty">https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty</a> ) |
| WEB-2 | Teesta dispute unresolved — Bangladesh and India would adjudicate water-allocation questions differently ( <a href="https://en.wikipedia.org/wiki/Teesta_Water_Dispute">https://en.wikipedia.org/wiki/Teesta_Water_Dispute</a> )                                                                                                                        |
| WEB-4 | Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ ( <a href="https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696">https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696</a> )         |

## Information Gaps

- Whether any individual MMLU item explicitly flags regional divergence in its answer key (none observed in sample)

## Requires Expert Verification

- Bangladeshi and Indian water-policy experts would be needed to systematically identify items whose correct answer differs by national perspective

## Remediation

**Gap 1: Four-option MCQ schema cannot encode legitimate regional divergence in correct answers (especially cross-border water/agricultural policy); moral\_scenarios encodes US 2020 norms structurally [DATASET-D1, D12].**

**Recommendation:** Exclude moral\_scenarios, human\_sexuality, human\_aging, and US-jurisdiction-specific configs from any deployment-relevant scoring. For the supplementary regional benchmark, design schema permitting parallel-correct-answer annotation flagged by jurisdiction (Bangladesh / India / state-level) for water-policy and agricultural-policy items.

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|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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